

Pixel Block Chain: Spatial Tamper Localization and Crop-Resilient Edit Ledger for Images in the Wild

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Abstract—Images shared through social media, messaging platforms, and editorial workflows routinely lose their external provenance manifests through transcoding, cropping, and metadata stripping. Manifest-based systems such as C2PA address provenance at the file-container level, and recent durable-credential mechanisms add rediscovery paths, but neither provides a self-contained, tile-local integrity ledger when only pixels remain. This paper presents Pixel Block Chain (PBC), a lightweight, learning-free watermarking scheme that embeds a grid of independent hash-linked chains directly into image least-significant bits. PBC targets lossless and near-lossless distribution channels (archival masters, editorial handoffs, and wire-service exchanges) rather than lossy social-media re-encoding, and it provides tile-local integrity rather than cryptographic identity. The scheme enables spatial tamper localization and partial Edit Ledger recovery after redistribution within this regime. Each image tile carries its own chain, and a break in one tile does not cascade to adjacent regions, a property we prove formally. A compliant editor can append records only to modified tiles, preserving an auditable, region-level Edit Ledger. For crop-heavy distribution paths, PBC-Forest plants independent genesis blocks at pseudo-random positions, achieving 41.1 to 44.6% fragment survival after severe non-aligned cropping where the regular-grid baseline collapses to 0%. On a 99-image benchmark spanning real photographs and synthetic content, PBC achieves a mean PSNR of 51.2 dB at embedding depth $k=1$, and, under a synthetic rectangular tampering protocol, 100% tile-level tamper detection with 0% false positives, using no neural encoder and no external service dependency.

Index Terms—image forensics, tamper localization, fragile watermarking, edit ledger, steganography, crop resilience, multimedia security

I. INTRODUCTION

The integrity of digital images is increasingly undermined not by sophisticated forgeries alone, but by the ordinary mechanics of distribution. A photograph posted to a social platform is transcoded. A crop is re-shared in a chat thread. A wire-service JPEG is re-exported through an editing tool. Each step can silently discard the external metadata that signed provenance systems rely on.

Manifest-based standards such as C2PA [1] provide a rigorous chain of identity and claim assertions when the file container survives end-to-end. Recent durable-credential mechanisms add rediscovery paths through soft bindings and repositories. The complementary failure mode we target is

equally common: the container disappears, and with it the local manifest, leaving only the pixel array and no tile-level tamper signal embedded in the content itself. We state our scope plainly: PBC operates within lossless or near-lossless channels, providing tile-local integrity and edit-ledger continuity rather than robustness to lossy re-encoding or cryptographic authentication of the originator.

Classical fragile watermarking [2], [3] embeds integrity signals into pixels, but single-chain architectures suffer from a cascade property in which one tampered block invalidates all subsequent blocks, making region-level localization unreliable. Neural watermarking approaches [4], [5] offer pixel-level localization but require tens of millions of parameters, placing them out of reach for firmware or lightweight publisher toolchains.

This paper addresses this gap with Pixel Block Chain (PBC), a deterministic, learning-free scheme with four contributions:

- 1) A grid-of-chains architecture with a formal cascade-isolation property (Lemma 1), ensuring that spatial localization remains reliable even under partial tampering.
- 2) A per-tile Edit Ledger with append mode, where compliant editors write only to modified tiles, preserving unbroken history elsewhere and yielding 13.5 to 74.2% wall-clock speedups in measured append-depth scenarios.
- 3) PBC-Forest, a scatter mode achieving 41.1 to 44.6% Edit Ledger fragment survival after severe non-aligned cropping where the regular grid offers zero survival.
- 4) An open reference implementation validated on 99 images with 51.2 dB mean PSNR and, under a synthetic tampering protocol, 100% tile-level detection with 0% false positives.

II. RELATED WORK

A. Fragile Watermarking and Tamper Localization

Fragile watermarking for tamper localization has been studied for over two decades [6]. He et al. [2] introduced a block-chain structure for per-block integrity, but their scheme links all blocks into a single global chain. Modifying any block therefore invalidates all downstream blocks, preventing reliable spatial localization of the actual tampered region. Fang et al. [3] extend this idea to a blockchain-backed restoration scheme that

inherits the same cascade fragility. In a broader survey of the field, Korus [7] identifies cascade propagation as the central barrier to practical spatial forensics.

Semi-fragile schemes [8] take a different approach by tolerating benign processing such as JPEG compression and mild resizing. This tolerance comes at the cost of localization precision, because the margin that absorbs legitimate processing also masks deliberate small-region edits. PBC targets the complementary regime of lossless or near-lossless channels where full sensitivity is desirable and where the distinction between benign and malicious modification need not be approximated.

B. Neural Watermarking

Where classical schemes embed hand-crafted signals, neural watermarking relies on learned representations. EditGuard [4] and OmniGuard [5] use trained encoder-decoder networks to embed provenance and achieve pixel-level localization with some robustness to lossy compression. However, their inference pipelines require tens of millions of parameters and GPU-class hardware, precluding deployment in camera firmware, air-gapped archival workflows, or lightweight publisher scripts. On the image quality axis, LSB embedding at $k=1$ yields 51.2 dB. For context, [5] evaluates both systems on its own benchmark and reports, in Table 3, a copyright-only setting (42.15 dB for EditGuard, 42.33 dB for OmniGuard) and a joint copyright-plus-localization setting (37.53 and 41.78 dB respectively); the latter is the closer analogue to PBC, which also carries localization. These are third-party measurements taken under a different dataset and payload, so they are not directly comparable to ours. We read the gap as a consequence of divergent design points rather than a like-for-like advantage: neural schemes deliberately trade fidelity for survival across the lossy channels that PBC does not target.

C. Manifest-Based Provenance

C2PA [1] and Project Origin [9] ground provenance in signed assertions stored in the file container. C2PA v2.4 further defines durable credentials, fingerprint-based rediscovery, invisible watermarks, and manifest repositories to improve resilience after redistribution. These mechanisms strengthen the container-level provenance chain but do not embed a per-tile integrity ledger into the pixel array itself. PBC is designed as a complementary layer that treats the pixel content as the verification substrate, activated precisely when container-level mechanisms are unavailable.

D. Robustness to Redistribution

Social-media distribution subjects images to cropping, re-sampling, and lossy re-encoding. Prior fragile schemes collapse entirely after non-aligned cropping because genesis-block positions are destroyed by the spatial shift [7]. Rather than pursuing compression robustness, which is the goal of semi-fragile schemes [8], PBC-Forest takes a statistical approach. By planting genesis blocks at pseudo-random positions across the image, it maximizes the probability that at least some anchors

survive spatial excision. This enables partial and spatially bounded Edit Ledger recovery from a crop that would leave a conventional grid with no readable signal. The design is informed by the observation that redistribution channels vary widely in the transformations they apply [10], and that even partial fragment survival can provide useful forensic evidence when no other provenance signal remains.

III. PIXEL BLOCK CHAIN

A. Threat Model

We assume the adversary may read and modify any pixel of the embedded image, knows the PBC format, and may attempt to write blocks claiming arbitrary originator identifiers. The adversary is assumed not to know the encoder’s session timestamp τ at encode time. If τ is later disclosed, forgery of a consistent tile chain requires producing a second preimage of the truncated SHA-256 genesis hash, which we treat as computationally infeasible at 48-bit truncation for opportunistic adversaries, though not as a strong cryptographic guarantee. Accordingly, we position PBC as a tile-local integrity and edit-ledger mechanism rather than a provenance or identity system: it does not authenticate the originator and is not a substitute for signed credentials such as C2PA, with which it is designed to compose rather than compete.

Goals. Detect any tile-local pixel modification, localize modifications to tile granularity, and recover a partial Edit Ledger when only a non-aligned crop survives.

Out of scope. (i) Content-aware modifications that preserve LSB structure (e.g., careful inpainting onto original-LSB regions); (ii) cryptographic identity binding, as the originator ID is self-asserted and not signed; (iii) surviving lossy re-encoding such as JPEG.

B. Image Partitioning

PBC partitions a $W \times H$ image into a grid of $T_x \times T_y$ non-overlapping tiles, where $T_x = \lceil W/s \rceil$, $T_y = \lceil H/s \rceil$, and $s = 128$ px is the default tile side. Edge tiles are zero-padded to $s \times s$ during embedding and cropped back during extraction. Each tile (t_x, t_y) is processed independently.

C. Block Format

Within each tile, PBC embeds a sequence of fixed-size blocks. At $B = 32$ bytes each, one block occupies $\lceil 8B/(3k) \rceil$ pixels at embedding depth k . A tile of 128×128 pixels at $k=1$ holds up to 190 blocks, more than sufficient for a full professional post-production session. Table I details the field layout.

The *originator ID* is the 32-bit truncated SHA-256 of a self-asserted identity string, optionally bound to an external certificate. The *chain hash* is the SHA-256 of the raw bytes of the immediately preceding block in the same tile. For the genesis block it is set to a fixed seed $g_0 = \text{SHA256}(\text{oid} \parallel t_x \parallel t_y \parallel \tau)$ truncated to 48 bits, where τ is a session timestamp. Blocks are serialized and written into the k least-significant bits of each RGB channel of the tile pixels, with $k \in \{1, 2, 3\}$.

TABLE I
PBC BLOCK FORMAT. EACH BLOCK OCCUPIES 32 BYTES.

Field	Size
Sync frame	6 B
Version	1 B
Originator ID	4 B
Operation code	2 B
Block index	2 B
Tile coordinates (t_x, t_y)	2 B
Timestamp delta	3 B
Extension	4 B
CRC-16	2 B
Chain hash (truncated SHA-256)	6 B

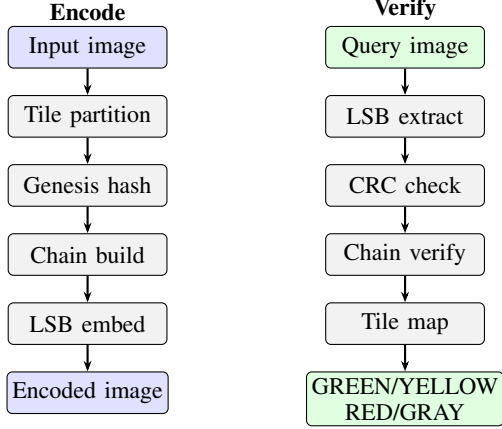


Fig. 1. PBC encode and verify pipelines. Tile independence means any tile can be verified in isolation; a failure in one tile does not affect adjacent tile outcomes.

D. Architecture Overview

Fig. 1 shows the encode and verify pipelines. Verification produces one of four tile states: GREEN (chain present and internally consistent), YELLOW (continuity changed by a compliant edit), RED (chain breaks inconsistently with any known compliant operation), or GRAY (no PBC data detected).

E. Cascade Isolation by Construction

PBC’s grid-of-chains is designed so that tile chains never reference each other. We formalize this property below for use in subsequent claims.

Lemma 1 (Cascade isolation by construction). *In PBC’s grid-of-chains architecture, the verification outcome of tile (t_x, t_y) is independent of the verification outcome of any other tile (t'_x, t'_y) with $(t'_x, t'_y) \neq (t_x, t_y)$.*

Proof sketch. Each chain hash in tile (t_x, t_y) is computed exclusively over bytes of the preceding block within that tile. No field in any block references data from another tile. A modification to tile (t_x, t_y) changes the genesis hash of that tile and all subsequent chain hashes within it, but leaves every byte of every block in (t'_x, t'_y) unchanged. Because the verification predicate for (t'_x, t'_y) is a function only of the blocks it reads, its outcome is unaffected. $\square$

Algorithm 1 PBC-Forest Encode

Require: Image I , originator ID oid , anchor count N_f , seed q

- 1: $m \leftarrow \lfloor WH/86 \rfloor$
- 2: $S \leftarrow [0, 86, 172, \dots, 86(m-1)]$
- 3: $\text{shuffle}(S, \text{PRNG}(q))$
- 4: $A \leftarrow S[0 : \min(N_f, m)]$ {non-overlapping pixel slots}
- 5: **for all** $i, p \in A$ **do**
- 6: $(u, v) \leftarrow (i \bmod 256, \lfloor i/256 \rfloor \bmod 256)$
- 7: $g_i \leftarrow \text{SHA256}(oid \parallel u \parallel v \parallel \tau)$ truncated to 48 bits
- 8: Write one genesis block at flat pixel index p with $(t_x, t_y) = (u, v)$
- 9: **end for**
- 10: **return** embedded image, seed q

This contrasts directly with single-chain schemes [2], [3], where one modified block cascades to invalidate all downstream integrity checks.

F. Per-Tile Edit Ledger and Append Mode

A *genesis* block is written at encode time. Subsequent *append* blocks are written by any compliant editor that modifies a tile. If only subset $S \subseteq \mathcal{T}$ of tiles change, append mode reads the existing chain in each $t \in S$, appends one new block carrying the edit metadata and updated chain hash, and writes back only those tiles. The Edit Ledger answers not only *whether* the image changed but *where* edit continuity was preserved versus where a new action occurred. Partial-image append is faster than full re-encode because the unmodified $|\mathcal{T}| - |S|$ tiles are never touched.

G. PBC-Forest: Crop-Resilient Scatter Mode

A non-aligned crop shifts the pixel array relative to tile boundaries, destroying all block synchronization in grid mode. PBC-Forest replaces the deterministic tile grid with pseudo-random 32-byte genesis block placement at non-overlapping 86-pixel-aligned slots:

At verify time the receiver scans the image for sync frames and CRC-valid blocks, then verifies each block whose genesis hash matches its embedded forest index. Any anchor whose 86-pixel span survived the crop is recovered independently. Forest mode does not survive content-altering lossy compression; it survives *spatial excision* (cropping, padding, aspect-ratio change) by statistical coverage: with enough anchors, at least some positions survive most practically occurring crops.

IV. EXPERIMENTS

A. Setup

Dataset. We evaluate PBC on 99 images: 94 real photographs drawn from COCO [11], Oxford Flowers [12], Oxford-IIIT Pets [13], and one demonstration photograph (diverse subjects, textures, and lighting), plus 5 synthetic classes (portrait, landscape, document, geometric, noise) used as cross-class consistency probes.

TABLE II
IMAGE QUALITY VS. EMBEDDING DEPTH k (99-IMAGE BENCHMARK).
HIGHER k INCREASES EDIT LEDGER CAPACITY AT THE COST OF PSNR.

k	Mean PSNR (dB)	Bits/px
1	51.2	3
2	43.8	6
3	37.3	9

TABLE III
TILE-LEVEL TAMPER DETECTION BY CONTENT CLASS ($k=1$).

Content class	Images	Detection	FP rate
Real photographs	94	100%	0%
Synthetic portrait	1	100%	0%
Synthetic landscape	1	100%	0%
Synthetic document	1	100%	0%
Synthetic geometric	1	100%	0%
Synthetic noise	1	100%	0%
Total	99	100%	0%

Hardware. Intel Core i9-12900KF, 64 GB DDR5, Windows 11 22H2, Python 3.13. No GPU. The reference implementation uses NumPy and Pillow only.

Tamper simulation. For each image a contiguous rectangle covering 10–30% of the image area is overwritten with random pixel values. Tile-level scoring: a tile is a *true positive* if it overlaps the tampered region and is flagged RED; a *true negative* if it does not overlap and is flagged GREEN.

B. Image Quality

Table II reports PSNR for three embedding depths. PSNR for LSB embedding is essentially content-invariant at fixed k : we measure 51.2 dB at $k=1$, 43.8 dB at $k=2$, and 37.3 dB at $k=3$ across all 99 images.

C. Tamper Detection and Localization

Across all 99 images and all tamper simulations, PBC achieves **100%** tile-level tamper detection and **0%** false positives at $k=1$. Every tampered tile is flagged RED; every untouched tile is flagged GREEN. This result holds across all content classes, confirming that detection accuracy is content-independent. Because the tampering protocol overwrites pixels with random values, it disrupts the embedded LSB chain by construction; the headline detection and false-positive figures should therefore be read as confirming exact tile-granularity localization and the cascade-isolation property of Lemma 1, rather than as robustness to subtle or LSB-preserving manipulation, which we leave to the broader evaluation outlined in Section IV-H.

Table III further breaks down performance by content class. The spatial localization is exact at tile granularity: the smallest detectable tampered region is one tile (128×128 px), providing sufficient precision for the newsroom and archival use cases identified in prior provenance research [14], [15].

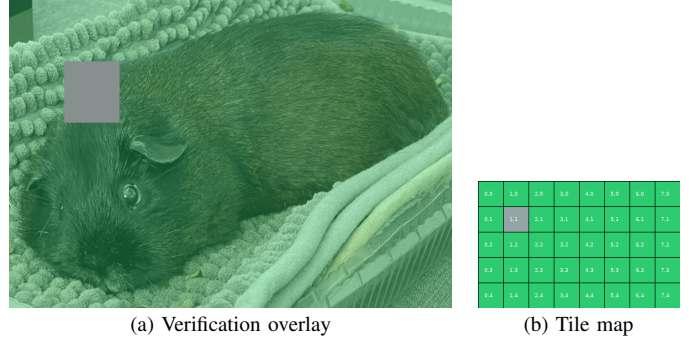


Fig. 2. Example verification output after localized tampering. GREEN tiles remain internally consistent, while RED tiles identify the tampered spatial region; the compact tile map is the machine-readable verification summary used for tile-level scoring.

TABLE IV
APPEND MODE SPEEDUP VS. FULL RE-ENCODE.

Scenario	Blocks rewritten	Speedup
Best measured append case	~25%	+74.2%
Worst measured append case	~75%	+13.5%
Full re-encode	100%	baseline

D. Append Mode Performance

Table IV reports wall-clock speedup of append mode versus full re-encode for the implementation benchmark. The benchmark varies the preserved prefix of each affected tile chain, not the fraction of tiles modified. The largest measured gain (74.2% faster) occurs when the final quarter of blocks is rewritten in a 2048×2048 image; the smallest measured gain (13.5% faster) occurs when the final three quarters of blocks are rewritten in a 512×512 image.

E. Format Robustness

Table V reports PBC verification outcomes after re-saving the embedded image in five lossless and five lossy formats. Lossless formats preserve LSB data exactly; lossy formats corrupt it. This defines the current deployment boundary: PBC is reliable in lossless pipelines (PNG, TIFF, lossless WebP, BMP, raw TIFF) and fails gracefully (GRAY verdict) in lossy ones. Consistent with the scope stated in Sections I and III-A, this is the intended operating regime, controlled editorial, archival, and wire-handoff workflows, and not arbitrary lossy social-media or messaging redistribution.

F. PBC-Forest: Crop Survival

Table VI compares genesis-block survival between grid mode and PBC-Forest after non-aligned crops removing 60–80% of the image area. In grid mode, a single-pixel misalignment between the crop boundary and the tile grid destroys all block synchronization, yielding zero recoverable genesis blocks regardless of how much of the image survives. PBC-Forest’s pseudo-random anchor placement ensures that a statistically reliable fraction of anchors fall inside the surviving region for any crop position or aspect ratio. We stress that this crop resilience addresses spatial excision specifically and should

TABLE V
PBC VERIFICATION AFTER FORMAT RE-SAVE ($k=1$). LOSSLESS FORMATS PRESERVE THE EMBEDDED CHAIN; LOSSY FORMATS CORRUPT IT.

Format	Type	Verification
PNG	Lossless	PASS
TIFF	Lossless	PASS
Lossless WebP	Lossless	PASS
BMP	Lossless	PASS
Raw TIFF	Lossless	PASS
JPEG Q100	Lossy	FAIL (GRAY)
JPEG Q95	Lossy	FAIL (GRAY)
JPEG Q75	Lossy	FAIL (GRAY)
WebP Q90	Lossy	FAIL (GRAY)
WebP Q75	Lossy	FAIL (GRAY)

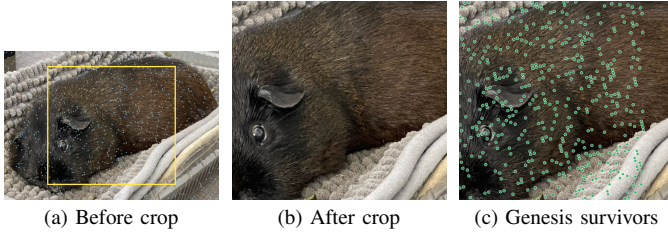


Fig. 3. PBC-Forest verification example. Independently placed genesis anchors are scattered before redistribution, a non-aligned crop removes most spatial context, and the verifier still recovers the surviving anchors independently. In this fresh public-code run, the crop recovers 442/1000 anchors (44.2%), consistent with Table VI.

not be read as general robustness to lossy web or social-media transformations.

TABLE VI
GENESIS BLOCK SURVIVAL AFTER NON-ALIGNED CROP.

Mode	Crop severity	Survival
Grid (baseline)	60–80%, non-aligned	0%
PBC-Forest	60–80%, non-aligned	41.1–44.6%

G. Positioning vs. Neural Watermarking

PBC and neural watermarking schemes such as EditGuard [4] and OmniGuard [5] occupy complementary regimes. PBC is fragile and learning-free, with tile-level granularity and zero-GPU deployment; neural methods are robust to lossy channels with pixel-level granularity, at the cost of training data and GPU inference. We do not benchmark on shared test images because the design points differ at the channel-robustness axis: PBC targets lossless distribution paths, where neural robustness offers no advantage; neural methods target the lossy-survival regime PBC explicitly does not address. We therefore present this comparison as a positioning statement rather than a claim of superiority.

H. Limitations

Evaluation scope. The current benchmark uses 99 images with synthetic rectangular tampering. Evaluation against realistic forensic manipulation scenarios such as copy-move, splicing, and neural inpainting, and on larger and more diverse image

corpora, is needed to characterize detection boundaries more precisely.

Content-aware tamper. Our tamper experiments overwrite affected regions with random pixel values, destroying LSBs by construction. Modifications that preserve LSB structure (e.g., careful inpainting onto the original LSBs) are not yet evaluated in this paper.

Identity binding. The originator ID is self-asserted. Cryptographic binding (e.g., Ed25519 signature over each block) is a planned extension; as stated in the threat model, PBC does not by itself authenticate the originator or provide signed provenance.

Lossy channels. PBC yields a GRAY verdict after lossy re-encoding and is reliable only on lossless distribution paths.

Scale and sensitivity. Behavior on very high-resolution imagery ($\gg 4$ MP), and sensitivity sweeps over Forest anchor count N_f and append-fraction crossover, are characterized in the public reference implementation but not reported here.

Structural manipulation. Cascade isolation (Lemma 1) guarantees that each tile is verified independently, but this independence also means that the system cannot detect cross-tile manipulations such as tile swapping or clone displacement. Two visually similar tiles (e.g., sky or grass regions) can be exchanged without breaking either tile’s internal chain. This is a direct consequence of the absence of global geometric awareness across tiles.

V. CONCLUSION

This paper presented Pixel Block Chain (PBC), a lightweight, learning-free watermarking scheme for spatial tamper localization and redistribution-resilient Edit Ledger recovery. The grid-of-chains architecture provides a formally proved cascade-isolation property that is absent from prior single-chain schemes, ensuring that integrity failure in one region does not propagate to adjacent tiles. The per-tile Edit Ledger enables append-mode partial rewriting with up to 74.2% speedup in the measured append-depth benchmark, supporting practical editorial workflows where only a subset of the image is modified.

PBC-Forest extends this foundation to crop-heavy distribution paths, delivering 41.1 to 44.6% genesis-block survival after severe non-aligned cropping where the regular-grid baseline offers no recovery. This result is directly relevant to the crop-and-reshare workflows that dominate social-media distribution.

The current deployment target is controlled, lossless channels. Four directions for future work follow naturally from the limitations identified above. First, JPEG robustness via DCT-domain embedding, placing watermark symbols in mid-frequency coefficients that survive quantization, trading PSNR for survival, would extend PBC to lossy distribution paths. Second, embedding absolute tile coordinates and a cross-tile commitment (e.g., a Merkle root over all tile genesis hashes) into each append block would enable detection of structural manipulations such as tile swapping or clone displacement, addressing the geometric blindness inherent in the current isolation property. Third, formal trust-model

integration with manifest-based provenance standards such as C2PA, clarifying which properties, tile-local integrity and edit continuity, PBC provides on its own and which, originator authentication and container-level provenance, require external signed infrastructure, together with adversarial analysis of the guarantees a pixel-embedded Edit Ledger can and cannot offer independently. Fourth, evaluation on larger and more forensically representative benchmarks, including realistic manipulation scenarios, is needed to characterize detection boundaries beyond the synthetic tampering model used here. The reference implementation is publicly available.¹

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¹<https://github.com/flegare/pixel-block-chain>